

Harthik Royal Mallichetty

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PROFESSIONAL SUMMARY

Data Engineer and May 2026 MSBA graduate with production experience designing and deploying end-to-end data pipelines, orchestration systems, and cloud infrastructure. Built Signal, a financial intelligence platform with an 8-module Python ingestion pipeline, dbt 3-layer transformation, Airflow orchestration, and PostgreSQL on GCP Cloud SQL covering 70 S&P 500 companies. Previously built a production ETL and ML system at Prolifics processing 50K+ rows/day, still running in 2025. Proficient in Python, SQL, dbt, Apache Airflow, PostgreSQL, GCP, Docker, and FastAPI.

EDUCATION

The University of Texas at Dallas, Richardson, Texas

Master of Science in Business Analytics and Artificial Intelligence – Scholar With Distinction

May 2026

GPA 3.81

GITAM University, Hyderabad, India

April 2024

Bachelor of Technology in Computer Science and Business Systems

GPA 3.73

PROFESSIONAL EXPERIENCE

Prolifics – Data & Analytics Intern | Hyderabad, India

Feb 2023 – Jan 2024

- Built SmartQE data pipeline solo: designed SQL workflows ingesting 100K+ PostgreSQL records, engineered data cleaning and transformation logic handling nulls and missing timestamps using same-sprint row logic, deployed output into Flask REST API integrated with company portal; system used by 200 developers daily, still in production 2025.
- Automated ETL workflows processing 50K+ rows/day using Python, reducing reporting cycle time by 40% and improving downstream accuracy by 25%; built Power BI dashboards tracking operational KPIs for Finance, DevOps, and QA teams.

HeiDigi – Digital Marketing Analyst Intern | Hyderabad, India

Mar 2022 – Aug 2022

- Built automated reporting pipeline pulling Google Ads performance data into Tableau dashboards, reducing manual reporting effort and surfacing KPIs that improved ROI by 20% and CTR by 18%.

PROJECTS

Signal – Financial Intelligence | Python, dbt, Airflow, PostgreSQL, GCP Cloud SQL, FastAPI, Docker

May 2026

- Designed 8-module Python ingestion pipeline pulling SEC EDGAR filings and Polygon.io market data: API clients with dual-key failover and rate limiting, XBRL parser with fallback chain and bank-specific logic, upsert loader that never creates duplicates on re-run, and failed ingestion logger so the pipeline never crashes on a single company failure.
- Architected dbt 3-layer transformation (Staging, Intermediate, Marts) computing 65 financial metrics per company across 70 S&P 500 companies; marts layer is the sole data source for the FastAPI layer, ensuring no raw table queries hit production. Orchestrated via 3 Airflow DAGs: initial load, daily 8-K monitor, weekly metrics refresh on GCP Cloud SQL.

Quant Risk Desk | Python, SQL, dbt, Apache Airflow, PostgreSQL, Docker, Tableau

April 2026

- Built production data pipeline via 9-task Airflow DAG ingesting 21K+ daily records across 11 S&P 500 sector ETFs and SPY; implemented 18 dbt data quality tests enforcing mathematical integrity constraints with zero violations, ensuring clean reliable data across all downstream layers.
- Designed 3-layer dbt transformation architecture with 11 SQL models computing rolling VaR/CVaR (95%/99%), Sharpe/Sortino, rolling Beta, and Macro Regime Engine using Fed Funds Rate, CPI, VIX, and yield curve signals; containerized full stack with Docker Compose for reproducible local development.

Global Supply Chain Intelligence | Python, SQL, XGBoost, LightGBM, SHAP, NetworkX, Plotly

Mar 2026

- Designed scalable data pipeline integrating 3 heterogeneous sources across 78K weekly records and 500 SKUs; built CUSUM and Mahalanobis anomaly detection achieving 100% disruption recall with 8-week lead time and XGBoost and LightGBM ensemble achieving Recall 70.9%.
- Executed Monte Carlo simulation with 10K runs per SKU across 3 geopolitical stress scenarios (baseline \$13.8M, moderate \$18.1M, severe \$19.1M); delivered 6-tab interactive Plotly dashboard for non-technical stakeholder review.

SKILLS

Programming and Databases: Python (Pandas, NumPy, scikit-learn, XGBoost, SHAP, SciPy, Plotly), SQL (CTEs, Window Functions, Self-Joins, Rolling Aggregations, Query Optimization), R, PostgreSQL, pgvector, DuckDB, MySQL, Git/GitHub

Data Engineering: dbt (Staging/Intermediate/Marts, custom tests, data quality enforcement), Apache Airflow (DAGs, batch orchestration), ETL Pipeline Automation, Data Validation and Quality Testing, Dimensional Modeling, Star Schema Design, Upsert Logic, Pipeline Fault Tolerance

Cloud and Infrastructure: GCP (Cloud Run, Cloud SQL), Docker, Docker Compose, GitHub Actions, FastAPI, Flask, n8n, SEC EDGAR API, Polygon.io

Streaming and Warehousing: Snowflake (familiar), Kafka (familiar), Databricks (familiar), Redis, Apache Parquet

ML and Statistics: XGBoost, LightGBM, TensorFlow, IsolationForest, CUSUM, Mahalanobis, A/B Testing, Hypothesis Testing, Time-Series Forecasting (ARIMA, Prophet), Monte Carlo Simulation, SHAP

ACTIVITIES

Cloud Analytics Lead – GDSC, GITAM Hyderabad

Apr 2022 – Apr 2024

- Built GCP and BigQuery pipelines analyzing behavioral data to quantify investment impact on college admission rates.